



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
11-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

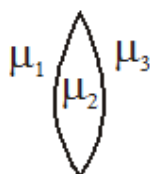
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. An equiconvex lens of refractive index μ_2 is used between two mediums of different refractive index μ_1 & μ_3 as shown. The system acts as a diverging lens if (considering paraxial rays)



(A) $\mu_1 = 1.2, \mu_2 = 2.4, \mu_3 = 3.8$

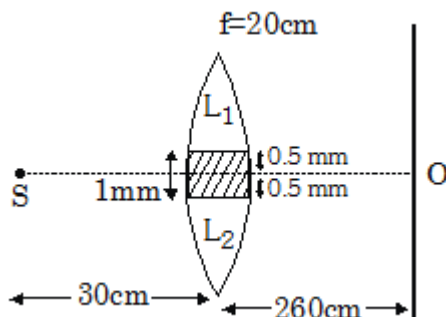
(B) $\mu_1 = 3, \mu_2 = 5/3, \mu_3 = 2$

(C) $\mu_1 = \frac{3}{2}, \mu_2 = \frac{7}{2}, \mu_3 = \frac{5}{2}$

(D) $\mu_1 = \frac{5}{3}, \mu_2 = \frac{10}{3}, \mu_3 = \frac{4}{3}$

Space for Rough Work

2. In billet split lens a thin convex lens of focal length 20 cm is cut in two halves L_1 and L_2 as shown and these halves are separated apart symmetrically from principal axis to a gap of 1 mm. Space between lens filled with opaque material. Assuming intensity from upper and lower half of lens near the central part of screen is I_0 each, the distance of source from lens is 30 cm and screen from lens is 260 cm (wavelength of source = 500 nm)



- (A) Separation between images formed by upper and lower half of lens is 3 mm.
- (B) The value of fringe width on screen is $\frac{1}{3}$ mm
- (C) Minimum distance from O where intensity is $4I_0$ is $\frac{1}{6}$ mm
- (D) Position of 3rd minima is at a distance of $\frac{5}{6}$ mm from O.

Space for Rough Work

3. A glass plate of refractive index μ_3 is coated with a thin layer of thickness t and refractive index μ_2 ($\mu_2 > \mu_3$). Light of wavelength λ travelling in air is incident normally on layer. It is partially reflected at the upper and the lower surfaces of layer. Choose correct option(s)

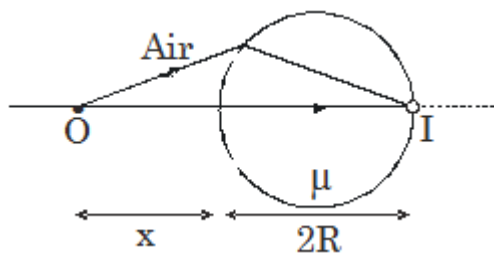
(A) The two reflected waves interfere constructively if $t = \frac{(n + \frac{1}{2}) \lambda}{2\mu_2}$.

(B) The two reflected waves interfere constructively if $t = \frac{(n + \frac{1}{2}) \lambda}{2\mu_3}$.

(C) The two reflected waves interfere destructively if $t = \frac{n\lambda}{2\mu_2}$

(D) The two reflected waves interfere destructively if $t = \frac{n\lambda}{2\mu_3}$

4. The figure shows a transparent sphere of radius R and refractive index μ . An object O is placed at a distance x from the pole of the first surface so that a real image is formed at the pole of the exactly opposite surface.



(A) If $x = 2R$, then the value of μ is 3

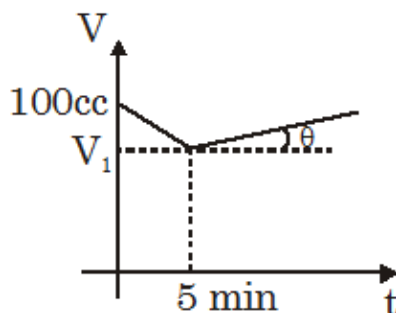
(B) If $x = \infty$, then the value of μ is 2

(C) If $x = 2R$, then the value of μ is 2

(D) If $x = \infty$, then the value of μ is 3

Space for Rough Work

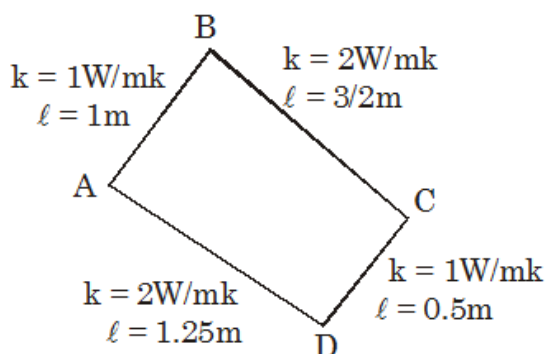
5. The maximum kinetic energy of photoelectrons ejected from a photometer when it is irradiated with radiation of wavelength 400 nm is 1 eV. If the threshold energy of the surface is 1.9 eV. (value of planck's constant is not known)
- (A) The maximum K.E. of photo electrons when it is irradiated with 500 nm photons will be 0.42 eV
- (B) The maximum K.E. in case (A) will be 0.49 eV.
- (C) The longest wavelength which will eject the photo electrons from the surface is nearly 610 nm.
- (D) Maximum K.E. will increase if the intensity of radiation is increased
6. A block of ice at 0°C was being heated at a constant rate. The volume of sample as a function of time is shown in the graph below. Take $\rho_{\text{ice}} = 0.9 \text{ gm/cc}$, $\rho_{\text{water}} = 1 \text{ gm/cc}$, $\gamma_{\text{water}} = 10^{-3} / ^\circ\text{C}$. Specific heat capacity of water = 1 cal/gm $^\circ\text{C}$, latent heat of fusion of ice = 80 cal/gm :-



- (A) $V_1 = 90 \text{ cc}$
- (B) $\frac{dH}{dt} = 24 \text{ cal/sec}$
- (C) $\tan \theta = 1.44 \text{ cc/min}$
- (D) Water will start boiling at 8.24 min.

Space for Rough Work

7. Consider 4 well lagged rods as shown. Point A is maintained at 100°C and C at 0°C . All rods have same cross sectional area :-



- (A) Temperature of junction D = $\frac{400}{7}^{\circ}\text{C}$
- (B) Temperature of junction B = $\frac{300}{7}^{\circ}\text{C}$.
- (C) Rate of heat flowing through AB is less than heat flowing through AD.
- (D) Rate of heat flowing through AB is more than rate of heat flowing through AD.
8. A certain mass of ideal gas undergoes a process given by $dU = \frac{dW}{2}$. If the molar heat capacity of the gas for this process is $\frac{15}{2}R$: (where dU = small change in internal energy; dW = small work done)
- (A) The gas is diatomic.
- (B) The gas is monoatomic.
- (C) The change in internal energy in the gas for an increase in temperature ΔT is $n \left(\frac{5R}{2} \Delta T \right)$.
- (D) Work done on the gas for an increase in temperature ΔT is $nR\Delta T$.

Space for Rough Work

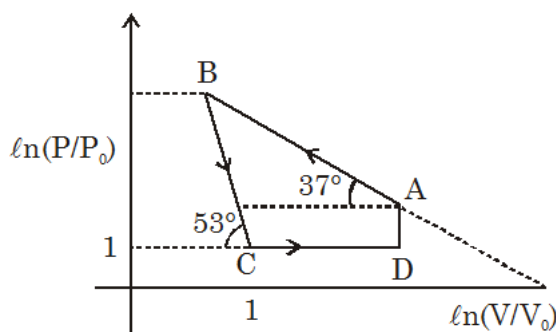
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln \left(\frac{P}{P_0} \right)$ versus $\ln \left(\frac{V}{V_0} \right)$ diagram.



Match the process.

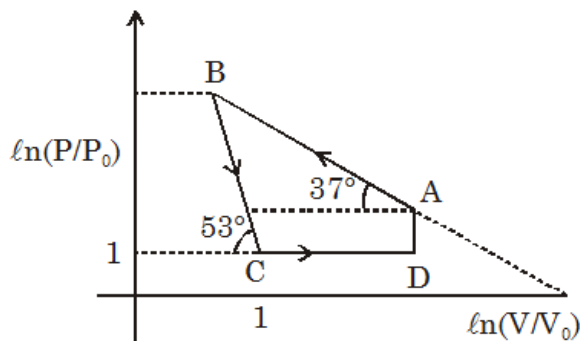
List-I		List-II	
(I)	AB	(P)	Polytropic $PV^m = \text{constant}$. ($m \neq \gamma$ or 1 or 0)
(II)	BC	(Q)	Adiabatic
(III)	CD	(R)	Isochoric
(IV)	DA	(S)	Isobaric
		(T)	Isothermal
		(U)	$P = a + bV$

- (A) I $\rightarrow$ P; II $\rightarrow$ U; III $\rightarrow$ R; IV $\rightarrow$ S
 (B) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ R
 (C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R
 (D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln\left(\frac{P}{P_0}\right)$ versus $\ln\left(\frac{V}{V_0}\right)$ diagram.



Match the process.

List-I		List-II	
(I)	$W_{A \rightarrow B}$	(P)	Positive
(II)	$Q_{B \rightarrow C}$	(Q)	Negative
(III)	ΔU_{CD}	(R)	Zero
(IV)	$Q_{D \rightarrow A}$	(S)	Can't be said
		(T)	Initially positive and negative after that
		(U)	Initially negative and positive after that

(A) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ P; IV $\rightarrow$ P

(B) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

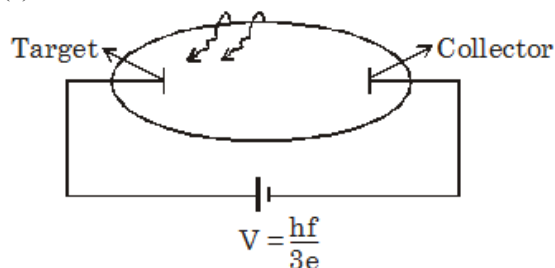
(D) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ P

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

According to Maxwell's theory, light is electromagnetic wave. Later on, Philip Lonard and his co-workers shared photoelectric effect. This could not be explained by wave theory. It could be explained only by Einstein's photoelectric effect equation. Although Maxwell's theory also predicts that a light beam exerts pressure on a surface, an easier explanation appears from photon theory.

In a photoelectric experiment, light of frequency f is incident on a metal target whose work function is $\phi = \frac{hf}{3}$. In list-I, KE of photoelectrons is mentioned at various location/instant and in list-II, the corresponding possible value(s).



List-I		List-II	
(I)	Maximum K.E. of photoelectrons just after emission	(P)	Zero
(II)	KE of photoelectrons just after emission	(Q)	$\frac{hf}{3}$
(III)	KE of photoelectrons when they are halfway between target & collector	(R)	$\frac{hf}{2}$
(IV)	KE of photoelectrons as they reach collector	(S)	$\frac{2hf}{3}$
		(T)	Can be equal to or more than energy of incident photon
		(U)	Must be less than the energy of incident photon

- (A) I $\rightarrow$ S,U; II $\rightarrow$ P,Q,R,S,U; III $\rightarrow$ P,Q,R,U; IV $\rightarrow$ P,Q,U
 (B) I $\rightarrow$ Q,R,U; II $\rightarrow$ P,R,S,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ P,Q,R,S,T
 (C) I $\rightarrow$ S,U; II $\rightarrow$ S,U; III $\rightarrow$ R,U; IV $\rightarrow$ Q,U
 (D) I $\rightarrow$ P,R,U; II $\rightarrow$ P,Q,R,S,T; III $\rightarrow$ P,R,S,T; IV $\rightarrow$ Q,R,S,U

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

According to Maxwell's theory, light is electromagnetic wave. Later on, Philip Lonard and his co-workers shared photoelectric effect. This could not be explained by wave theory. It could be explained only by Einstein's photoelectric effect equation. Although Maxwell's theory also predicts that a light beam exerts pressure on a surface, an easier explanation appears from photon theory.

List-I		List-II	
(I)	Bichromatic light source emitting unidirectional light	(P)	Some photons have same energy & momenta
(II)	Point source of white light emitting light uniformly in all directions.	(Q)	Some photons have different energy & momenta
(III)	Point source of monochromatic light emitting light uniformly in all directions	(R)	Some photons have same energy & different momenta
(IV)	Monochromatic laser light source emitting unidirectional light	(S)	Some photons have different energy & same momenta.
		(T)	There is no net force on the light source due to emitted photons.
		(U)	The emitted photons may exert a net force on the light source.

- (A) I $\rightarrow$ P,Q,U; II $\rightarrow$ Q,R,T; III $\rightarrow$ P,Q,R,S,T; IV $\rightarrow$ Q,U
 (B) I $\rightarrow$ P,Q,R,S,T; II $\rightarrow$ P,Q,R,S,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ P,T
 (C) I $\rightarrow$ P,Q,R,S,U; II $\rightarrow$ P,Q,R,T; III $\rightarrow$ P,S,T; IV $\rightarrow$ P,R,U
 (D) I $\rightarrow$ P,Q,U; II $\rightarrow$ P,Q,R,T; III $\rightarrow$ P,R,T; IV $\rightarrow$ P,U

Space for Rough Work

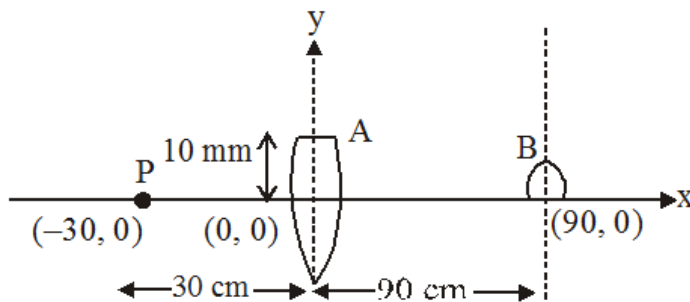
SECTION-II : (Maximum Marks: 18)

- This section contains **EIGHT (08)** questions, from which attempt any Six questions. In case if more than six questions will be attempted, then only first six questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A thin convex lens having focal length of 20 cm is cut into two parts (A and B) 10 mm above the principle axis. The lower part 'A' is placed with its optical centre at origin and upper part B at (90 cm, 0) as shown in the figure. A point object P is placed at (-30 cm, 0). If the distance of final image from x-axis is $\frac{N}{10}$ mm, then find the value of N (Assuming paraxial ray approximation to remain valid)



2. An electron moving along x-axis is confined between two walls at $x = 0$ and at $x = L$. There is no loss of energy during collision between electron and the wall. Initially the electron is in quantum state $n = 2$, and it jumps to quantum state $n = 1$. The emitted photon is incident on a metal having work function equal to energy of electron in quantum state $n = 1$. If maximum possible kinetic energy of emitted photoelectron is x times the energy of electron in quantum state $n = 1$, then find x .
3. Geiger counter reading of a radioactive sample is initially 6800 counts per min. The same sample gives a reading of 425 counts per min 10 hrs later. The sample's half life of α hr. Fill 2α .

Space for Rough Work

4. A radioactive sample contains two radio nucleoids A and B having decay constant $\lambda \text{ hr}^{-1}$ and $2\lambda \text{ hr}^{-1}$. Initially 25% of total decay comes from A. How long (in hr) will it take before 75% of total decay comes from A. [Take $\lambda = \ln 3$]
5. Neon 23 undergoes β decay
 ${}_{10}\text{Ne}^{23} \rightarrow {}_{11}\text{Na}^{23} + {}_{-1}^0\beta + \bar{\nu}$
 Atomic mass of ${}^{23}\text{Ne}$ and ${}^{23}\text{Na}$ are 22.9945 u and 22.9898u respectively. Mass of electron is $0.51 \frac{\text{MeV}}{c^2}$.
 The maximum kinetic energy of β^- particle is _____ MeV. (Take $1\text{u} = 931.5 \text{ MeV}/c^2$)
6. The de-broglie wavelength of electrons striking the cathode to produce x rays is equal $\frac{1}{100}$ of minimum wavelength of x ray produced. What is the voltage across x ray tube (in volt).
7. A cubical block of coefficient of linear expansion α_s is submerged partially inside a liquid of coefficient of volume expansion γ_ℓ . On increasing the temperature of the system by ΔT , the height of the cube inside the liquid remains unchanged. Find $\gamma_\ell/\alpha_s = ?$
8. A solid copper sphere ($e = 1$, density ρ and specific heat capacity c) of radius r at an initial temperature 200 K is suspended inside a chamber whose walls are at almost 0K. The time required (in μs) for the temperature of the sphere to drop to 100 K is $\frac{n}{72} \frac{r\rho c}{\sigma}$. Find the value of n. (All symbols have their usual meaning)

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

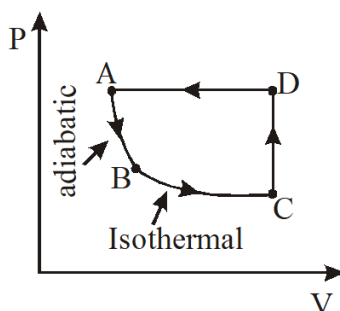
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Select the correct option(s).

- (A) A cycle is drawn on PV diagram in which there is one isothermal expansion & two adiabats (expansion & compression), then engine is 100 % efficiency.
- (B) A cycle is drawn on PV diagram in which there is one isothermal expansion & two adiabats (expansion & compression), then engine is not possible
- (C) Carnot engine has same efficiency if ideal gas or real gas is taken as fluid.
- (D) First law of thermodynamic for open system is dependent on mass exchange and enthalpy.

Space for Rough Work

2. 1 moles of an ideal gas, $[\gamma = 1.5]$ is taken through a series of reversible processes :



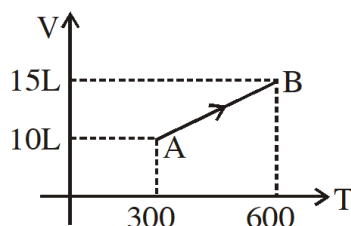
Information - I The temperature at state 'A' is 300 K.

Information - II The reversible isothermal expansion from B to C doubles the volume

Information - III The entropy change of the system from C to D is $4 \ln 16$ Cal/ K.

Select the correct option (s) using above information.

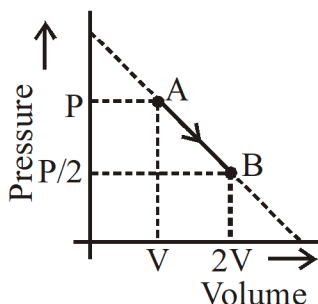
- (A) The entropy change of the system from D to A is $-3 R \ln 8$
- (B) The temperature at point D is 2400 K
- (C) The work done from A to B is -600 Cal.
- (D) The work done from B to C is $-300 \ln 2$ Cal.
3. If one mole monoatomic ideal gas was taken through process AB as shown in figure, then select correct option(s) Given : $\ln 1.5 = 0.4$



- (A) $w_{AB} = -1496.52$ J
- (B) $q_{AB} = 5237.82$ J
- (C) $\Delta H_{AB} = 3741.3$ J
- (D) ΔS_{AB} is + ve

Space for Rough Work

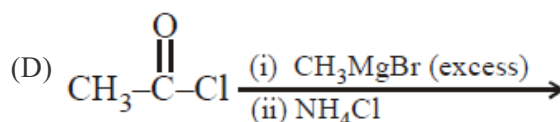
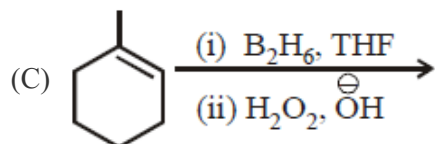
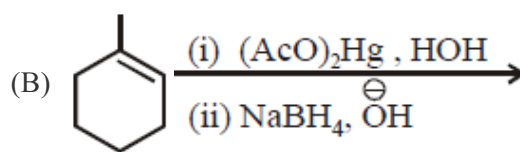
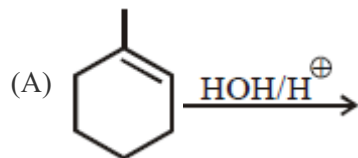
4. An ideal gas is taken from state A (Pressure P , Volume V) to the state B (Pressure $P/2$, Volume $2V$) along a straight line path in PV diagram as shown in the adjacent figure. Select the correct statement(s) among the following.



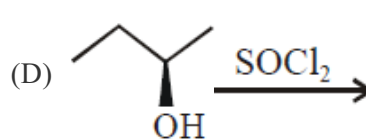
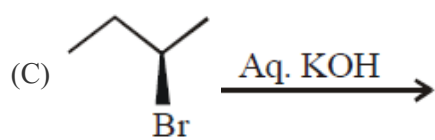
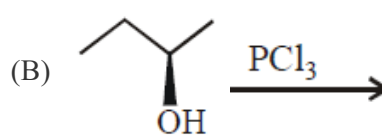
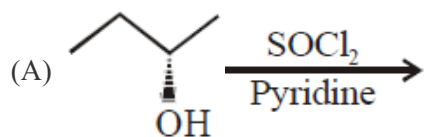
- (A) The work done by gas in the process A to B exceeds the work that would be done by it if the system were taken from A to B along the isotherm.
- (B) In the T - V diagram, the path AB become part of parabola
- (C) In the P - T diagram, the path AB becomes a part of hyperbola.
- (D) In going from A to B, the temperature T of the gas first increases to a maximum value then decreases.
5. Select the correct statement(s) for the complex $[\text{Co}(\text{en})(\text{NH}_3)_2(\text{CN})_2]\text{ClO}_4$:
- (A) Total number of stereo-isomers = 3, and four Co-N linkage
- (B) Number of geometrical isomers = 3, and one chelate ring
- (C) Number of optically inactive isomers = 2, and diamagnetic nature
- (D) Number of optically active isomers = 2, and d^2sp^3 hybridisation of central metal

Space for Rough Work

6. Decomposition of H_2O_2 is accelerated by
- (A) Acetanilide (B) Finely divided metals
- (C) Urea (D) Dust
7. In which of the following reaction(s), 2° alcohol is not obtained as major product.



8. Reaction(s) in which inversion of configuration takes place is/are?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9.

List-I Complex Compound		List-II Properties	
(I)	$[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$	(P)	d^2sp^3 hybridisation
(II)	$[\text{Co}(\text{CN})_2(\text{NH}_3)_4]\text{OC}_2\text{H}_5$	(Q)	Ionisation isomerism
(III)	$[\text{IrCl}_6]^{3-}$	(R)	$\mu = 2.83$ B.M.
(IV)	$[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$	(S)	$\Delta_0 < P$
		(T)	sp^3d^2 hybridisation
		(U)	geometrical isomerism

Which of the following option has the **correct** combination considering list I and list II.

- (A) I $\rightarrow$ P (B) II $\rightarrow$ R (C) I $\rightarrow$ S (D) II $\rightarrow$ T

Space for Rough Work

10.

List-I Complex Compound		List-II Properties	
(I)	$[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$	(P)	d^2sp^3 hybridisation
(II)	$[\text{Co}(\text{CN})_2(\text{NH}_3)_4]\text{OC}_2\text{H}_5$	(Q)	Ionisation isomerism
(III)	$[\text{IrCl}_6]^{3-}$	(R)	$\mu = 2.83$ B.M.
(IV)	$[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$	(S)	$\Delta_0 < P$
		(T)	sp^3d^2 hybridisation
		(U)	geometrical isomerism

Which of the following option has the **INCORRECT** combination considering list I and list II.

 (A) III $\rightarrow$ P

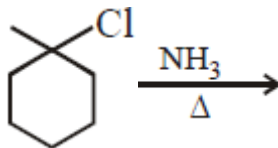
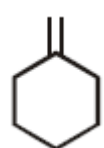
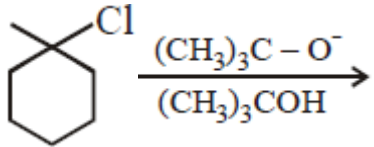
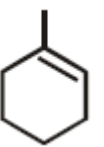
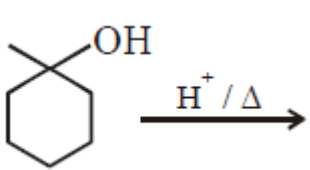
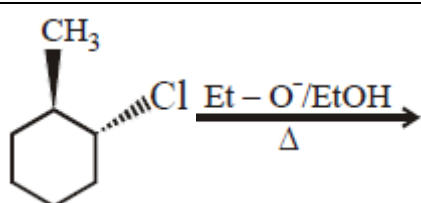
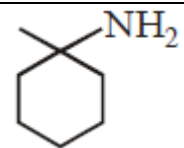
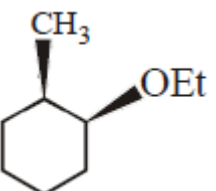
 (B) III $\rightarrow$ R

 (C) IV $\rightarrow$ Q

 (D) IV $\rightarrow$ U

 Space for Rough Work

11.

List-I Reaction		List-II Major product	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

Which of the following set of combination is **correct**?

(A) I $\rightarrow$ T

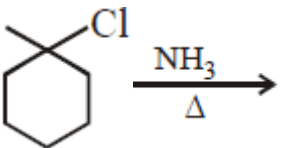
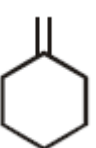
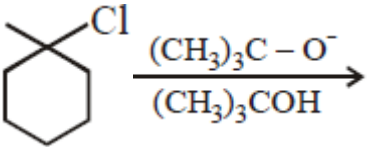
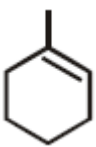
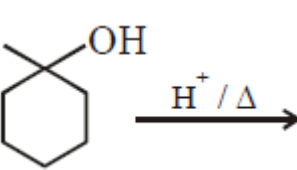
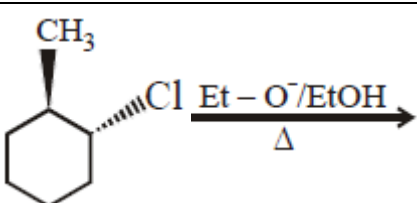
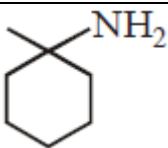
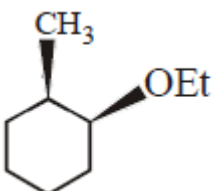
(B) IV $\rightarrow$ U

(C) III $\rightarrow$ P

(D) IV $\rightarrow$ R

Space for Rough Work

12.

List-I Reaction		List-II Major product	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

Which of the following set of combination is **incorrect**?

(A) I $\rightarrow$ Q

(B) II $\rightarrow$ P

(C) III $\rightarrow$ R

(D) IV $\rightarrow$ R

Space for Rough Work

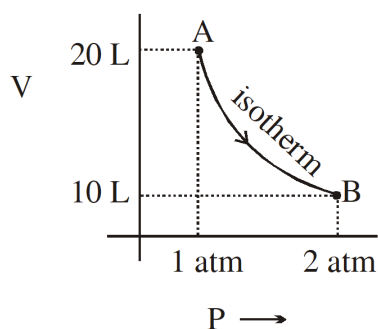
SECTION-II : (Maximum Marks: 18)

- This section contains **EIGHT (08)** questions, from which attempt any Six questions. In case if more than six questions will be attempted, then only first six questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. For an ionic solid MX_2 , where X is monovalent, the enthalpy of formation of the solid from $M(s)$ and $X_2(g)$ is 1.5 times the electron gain enthalpy of $X(g)$. The first and second ionisation enthalpies of the metal (M) are 1.2 and 2.8 times of the enthalpy of sublimation of $M(s)$. The bond dissociation enthalpy of $X_2(g)$ is 0.8 times the first ionisation enthalpy of metal and it is also equal to one-fifth of the magnitude of lattice enthalpy of MX_2 . If the electron gain enthalpy of $X(g)$ is -96 Kcal/mol , the answer the enthalpy of sublimation of metal (M) in Kcal/mol .
2. 1 mole of gas X is present in a closed adiabatic vessel fitted with a movable frictionless piston. The initial temperature of gas X is 300 K. The vessel is maintained at constant pressure of 1 atm. Keeping the pressure constant at 1 atm the reaction $(3X(g) \rightarrow 2Y(g))$; $\Delta H = -30 \text{ kJ/mol}$ is started with the help of negligible amount of electric energy. If finally 75 mole % of X undergone reaction at constant pressure of 1 atm, find the final temperature (in K) of reaction vessel.
Given : $C_{p,m(X)} = 40 \text{ J/K mole}$; $C_{p,m(Y)} = 30 \text{ J/K mole}$
3. Calculate ΔG (in L atm) from the graph of an ideal gas undergoing process AB. [$\ln 2 = 0.7$]

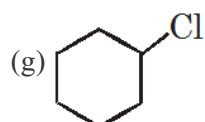
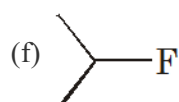
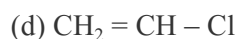
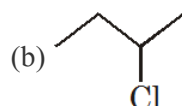


Space for Rough Work

4. If 'a', 'b' and 'c' are monodentate ligands then how many of the following complexes of a metal 'M' will show geometrical isomerism. [consider square planar geometry in coordination number 4 and octahedral geometry in coordination number 6]

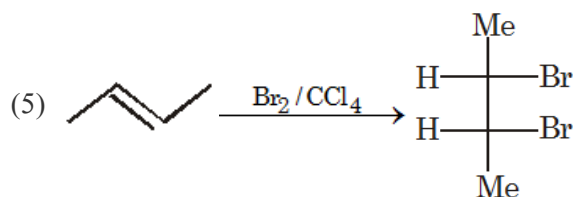
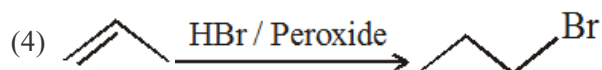
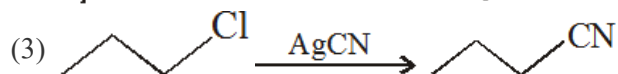
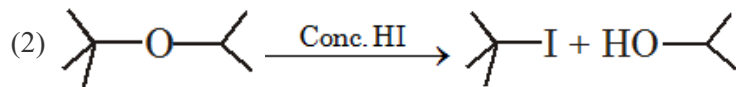
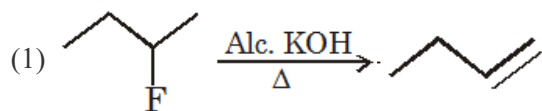
Ma_3b , Ma_2b_2 , Ma_6 , Ma_5b , Ma_4b_2 , Ma_3b_3 , Ma_4 , Ma_4bc , Ma_2bc , $\text{Ma}_3\text{b}_2\text{c}$, Mb_5c

5. What is the spin only magnetic moment (in Bohr Magnetons) of the complex $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$?
6. How many of the following are less reactive than $\text{Me}-\text{Cl}$ in an S_N^2 mechanism reaction



Space for Rough Work

7. Identify total number of structural isomers for $C_4H_{10}O$ (Alcohol & ether) :
8. Total number of reactions correctly match with their major products are ?



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
1. If $|Z| = \min. \{|Z - 1|, |Z + 1|\}$ where 'Z' is a complex no. and f be a one-one function from $\{a, b, c\}$ to $\{1, 2, 3\}$ and $f(a) = 1$ is false, $f(b) \neq 1$ is false and $f(c) \neq 2$ is true, then $|Z + \bar{Z}| =$

- (A) $f(a)$ (B) $f(c)$ (C) $\frac{1}{2} f(a)$ (D) $f(b)$
-

Space for Rough Work

2. Consider a curve $C : y = \sqrt{\frac{3x^2 - 21}{7}}$ and a line L of slope 1, which is tangent to curve C at point A , then answer the following question
- (A) Equation of L is $y = x + 2$
- (B) Equation of L is $y = x - 2$
- (C) Equation of circle passing through origin and touches the curve C at A is $x^2 + y^2 + x - 41y = 0$
- (D) Equation of circle passing through origin and touches the curve C at A is $4x^2 + 4y^2 + x - 41y = 0$
3. If a tangent of slope equal to $\frac{1}{3}$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$) is normal to the circle $x^2 + y^2 + 2x + 2y + 1 = 0$, then
- (A) Maximum area of ellipse is $\frac{2\pi}{3}$
- (B) $a \in \left(\sqrt{\frac{2}{5}}, 2\right)$
- (C) maximum area of ellipse is π
- (D) $a \in \left(\frac{2}{3}, \frac{3}{2}\right)$

Space for Rough Work

4. If z_1, z_2 are two non-zero complex numbers such that $|19z_1 - 31z_2|^2 = |19z_1|^2 + |31z_2|^2$ then
- (A) $\arg\left(\frac{z_2}{z_1}\right) = \pm \frac{\pi}{2}$ (B) $\frac{z_2}{z_1}$ is purely real
- (C) $\frac{z_2}{z_1}$ is purely imaginary (D) $z_1 \bar{z}_2 + z_2 \bar{z}_1 = \log_3 (\sin^2 \theta + \cos^2 \theta)$
5. Mr. X travels from Delhi to Muscat with change of airplanes in Chennai, Dubai and Sharjah respectively. He has one piece of luggage. At each stop the luggage is transferred from one airplane to another. At the airport in Delhi there is a probability of $\frac{1}{20}$ that luggage is not placed in right plane. This probability is $\frac{1}{10}$ at airport in Chennai, $\frac{1}{5}$ at airport in Dubai and $\frac{1}{10}$ at airport in Sharjah
- (A) The probability that Mr. X luggage does not reach Sharjah with him is 0.316
- (B) If Mr. X luggage does not reach muscat with him, then the probability that it was lost at airport of Dubai is $\frac{855}{1922}$
- (C) The probability that Mr. X luggage does not reach Sharjah with him 0.512
- (D) If Mr. X luggage does not reach muscat with him, then the probability that it was lost at airport of Dubai is $\frac{715}{1922}$

Space for Rough Work

6. If A and B are two matrices of order '3' such that $2A + 3BB^T = I$ and $B^{-1} = A^T$, then identify which of the following statement(s) is(are) correct ?
- (A) the value of $\text{Tr}(A^{-1} - 3B^3 - BA + I) = 6$ (B) the value of $\text{Tr}(A^{-1} - 3B^3 + BA) = 9$
 (C) the value of $\det(A^2 - 2A^3) = 27$ (D) the value of $\det(A^{-1} - 3B^3 + BA) = 81$
7. If $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, where $\alpha \in \mathbb{R}$; then choose the correct option(s)
- (A) $(A_\alpha)^{-1} = A_{(-\alpha)}$
 (B) $\det \left(\left(\frac{1}{2}(A_\alpha) \right)^{-1} \right) = 4$
 (C) $A_\pi A_{\frac{3\pi}{2}} A_{\frac{5\pi}{4}} A_{\frac{7\pi}{8}} A_{\frac{9\pi}{16}} \dots$ upto infinite terms $= I_2$
 (D) $\text{adj}(\text{adj}(\text{adj} A_\alpha)) = A_{(-\alpha)}$
 (where I_2 denotes the identity matrix of order 2)
8. Let a variable circle touches a line, which is tangent to the parabola $y^2 = 4x$ at (4, 4) and variable circle passes through focus of the parabola $y^2 = 4x$. If locus of its centre of the circle is conic 'C', then
- (A) directrix of conic 'C' is $x - 2y + 4 = 0$ (B) focus of conic 'C' is (4, 0)
 (C) latus rectum of conic 'C' is $2\sqrt{5}$ (D) axis of the conic 'C' is $2x + y - 2 = 0$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph
 India and Australia play a series of 'n' one day matches and probability that India wins a match against Australia is $\frac{1}{2}$. (assume that no match ends in a draw)

List-I		List-II	
(I)	If 'n' is not fixed and series ends when any one of the team completes its 4 th win then probability that India wins the series is	(P)	$\frac{4}{2^7}$
(II)	If n = 7 then probability that India wins atleast three consecutive matches is	(Q)	$\frac{47}{2^7}$
(III)	For n = 7; probability that India wins the series through exactly four wins that are consecutive is	(R)	$\frac{1}{2}$
(IV)	For n = 7; probability that India doesn't win two consecutive matches is	(S)	$\frac{9}{2^7}$
		(T)	$\frac{17}{2^6}$
		(U)	$\frac{35}{2^7}$

Which of the following option has the **CORRECT** combination considering list I and list II.

- (A) I → U (B) III → S (C) II → Q (D) IV → S

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph
India and Australia play a series of 'n' one day matches and probability that India wins a match against Australia is $\frac{1}{2}$. (assume that no match ends in a draw)

List-I		List-II	
(I)	If 'n' is not fixed and series ends when any one of the team completes its 4 th win then probability that India wins the series is	(P)	$\frac{4}{2^7}$
(II)	If n = 7 then probability that India wins atleast three consecutive matches is	(Q)	$\frac{47}{2^7}$
(III)	For n = 7; probability that India wins the series through exactly four wins that are consecutive is	(R)	$\frac{1}{2}$
(IV)	For n = 7; probability that India doesn't win two consecutive matches is	(S)	$\frac{9}{2^7}$
		(T)	$\frac{17}{2^6}$
		(U)	$\frac{35}{2^7}$

Which of the following option has the **INCORRECT** combination considering list I and list II.

- (A) I → R (B) IV → U (C) III → P (D) IV → T

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I		List-II	
(I)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i (i = 1, 2, 3, 4, 5)$ are digits and satisfying $d_1 < d_2 \leq d_3 < d_4 \leq d_5$, is	(P)	$^{10}C_4$
(II)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i (i = 1, 2, 3, 4, 5)$ are digits satisfying $d_1 > d_2 \geq d_3 > d_4 > d_5$, is	(Q)	$^{11}C_4$
(III)	Pankaj and Prateek play a unique game series in a chess tournament. They decide to play on till one of them wins 5 matches. If each match end only in win or loss. Number of ways in which series can be won by either of them, is	(R)	$^{11}C_6$
(IV)	A badminton team has to be selected comprising of 5 students out of 10 students for inter school tournament. Number of ways this can be done if a particular player is to be always included or always excluded from the team, is	(S)	9C_4
		(T)	$^{12}C_6$
		(U)	$^{10}C_5$

Which of the following option has the **CORRECT** combination considering list I and list II.

- (A) $I \rightarrow Q$ (B) $II \rightarrow T$ (C) $III \rightarrow U$ (D) $IV \rightarrow P$

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I		List-II	
(I)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i (i = 1,2,3,4,5)$ are digits and satisfying $d_1 < d_2 \leq d_3 < d_4 \leq d_5$, is	(P)	$^{10}C_4$
(II)	Number of five digit numbers of the form $d_1d_2d_3d_4d_5$ where $d_i (i = 1,2,3,4,5)$ are digits satisfying $d_1 > d_2 \geq d_3 > d_4 > d_5$, is	(Q)	$^{11}C_4$
(III)	Pankaj and Prateek play a unique game series in a chess tournament. They decide to play on till one of them wins 5 matches. If each match end only in win or loss. Number of ways in which series can be won by either of them, is	(R)	$^{11}C_6$
(IV)	A badminton team has to be selected comprising of 5 students out of 10 students for inter school tournament. Number of ways this can be done if a particular player is to be always included or always excluded from the team, is	(S)	9C_4
		(T)	$^{12}C_6$
		(U)	$^{10}C_5$

Which of the following option has the **INCORRECT** combination considering list I and list II.

- (A) $I \rightarrow R$ (B) $IV \rightarrow P$ (C) $II \rightarrow R$ (D) $IV \rightarrow U$

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SECTION-II : (Maximum Marks: 18)

- This section contains **EIGHT (08)** questions, from which attempt any Six questions. In case if more than six questions will be attempted, then only first six questions will be considered for marking.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let (x, y) be a variable point on the curve $x^2 + 4y^2 - 4x + 8y + 4 = 0$. If $\lambda = \frac{\max(x^2 + y^2 + 4x + 2y + 5)}{\min(x^2 + y^2 + 4x + 2y + 5)}$, then $\frac{\lambda}{5}$ equals
 2. If nine numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are put into a matrix of order 3×3 randomly so that each number occur exactly once. If the probability that the sum of the numbers in atleast one row is greater than 21 is $\frac{1}{p}$, then the value of $\frac{p}{4}$ equals
 3. Three digit numbers xyz are formed with digits from the set $\{0, 1, 2, \dots, 9\}$ such that $x \leq y \leq z$. If the number of such number is λ , then $\frac{\lambda}{100}$ equals
 4. Let two distinct numbers a and b are chosen at random from the set $\{1, 2, 3, \dots, 50\}$. If the probability that $(4^a + 4^b + 3)$ is divisible by 5 is λ , then 7λ equals
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5. Let a, b, c be three real numbers such that $a \in \left[0, \frac{\pi}{2}\right]$, $b \in \mathbb{R} - \{0\}$ and $c \in \mathbb{R}$. If

$$|(\sin a + \cos a) - \sqrt{2}| + \left|b + \frac{1}{b} - 2\right| + |c| \leq 0 \text{ and } D = \begin{vmatrix} \tan a & \cos^2 a & 1 \\ b+2 & b^3 & \frac{1}{b} \\ \ln(c+1) & c+5 & c^3+2 \end{vmatrix}, \text{ then value of } \frac{D}{4}$$

equals

6. Let $a, b, c, x, y, z \in \mathbb{R}$ such that

$$ax + by + cz = 1$$

$$bx + cy + az = 2$$

$$cx + ay + bz = 3.$$

If the value of $(a^3 + b^3 + c^3 - 3abc)(x^3 + y^3 + z^3 - 3xyz) = \lambda$, then $\frac{\lambda}{5}$ equals

7. Let ω be a non real complex number satisfying $\omega^{23} = 1$. If $N = \sum_{k=1}^{22} \frac{1}{1 + \omega^{8k} + \omega^{16k}}$ then $\frac{N}{6}$ equals

8. PQ and PR are two tangents to a parabola $y^2 = 4ax$ and tangent to parabola at a third point S cuts these tangents at A and B respectively then value of $\frac{1}{4} \left(\frac{PA}{PQ} + \frac{PB}{PR} \right)$ is (S lies on the arc QR of the parabola)

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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